

Supplementary Material

Sub-national Case Study: Germany (Kreis level)

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This supplement replicates the paper’s metric comparison and driver analysis at the **sub-national** level, using Germany as a case study. It serves two purposes: to show that the instability of the official LCRPGR indicator, and the usability of the Built-up per Capita Rate (BpCR), persist at a finer spatial scale; and to bring in **internal (within-country) migration**, which the national analysis cannot observe.

1 Data and Method

The dependent variable is built exactly as in the main paper, but for German districts (*Kreise*) rather than countries. Urban built-up area V and urban population P come from the GHSL grids (urban-masked at $\text{GHS-SMOD} \geq 21$), aggregated to the GAUL level-2 units for Germany at the five-year epochs 1985–2020; BpCR, the log-LCRPGR ratio, and urban density are then computed from these totals just as nationally. Two controls are taken from the German regional database INKAR (Federal Institute for Research on Building, Urban Affairs and Spatial Development, BBSR): GDP per capita (workplace-based) and the **internal net-migration rate** (*Binnenwanderungssaldo*, per 1,000 inhabitants), the latter being the within-country flow that the national UN measure omits. Urban population share is the GHSL urban population over the INKAR total district population. GAUL districts are matched to INKAR *Kreise* by name and type (city vs. rural district); 317 districts match cleanly and enter the regressions, over the epochs 1995–2020 for which the INKAR controls are available (1902 district-periods). All specifications carry district and period fixed effects with standard errors clustered by district; the dynamic models add the lagged dependent variable and use Arellano-Bond and Blundell-Bond GMM, as in the main paper.

2 The Metric Breaks Down Locally Too

Even at the district level the contrast is stark. Across the 1902 district-periods the log-LCRPGR ratio has a standard deviation roughly **6,708 times** larger than that of BpCR: at this fine scale many districts have near-zero population growth, so the ratio diverges even more violently than nationally, whereas BpCR stays in a tight, interpretable band.

Table 1: Static two-way fixed-effects models at the German Kreis level (1995-2020); log-LCRPGR ratio vs. BpCR, same sample and controls.

Variables	LCRPGR (log)	BpCR
ln(Urban Density)	+15.1200 (16.9767)	-0.0141*** (0.0035)
Urban Pop. Share	+4.2539 (42.9667)	+0.0162* (0.0087)
ln(GDP per capita)	+0.1818 (11.1630)	-0.0058*** (0.0018)
Internal net migration (per 1,000)	+0.0924 (0.1292)	-0.0000 (0.0000)
Observations	1,902	1,902
Within R ²	0.0005	0.0355
District FE	Yes	Yes
Period FE	Yes	Yes

*** p<0.01, ** p<0.05, * p<0.1.

Standard errors clustered by district, in parentheses.

As nationally, BpCR yields interpretable, significant coefficients while LCRPGR does not. Income enters BpCR negative and significant: within German districts, getting richer is associated with *less* built-up land per resident, the same densification result found across countries. The LCRPGR column returns enormous, mostly insignificant coefficients dominated by its unstable tails.

3 Dynamic Models and Internal Migration

Table 2: Dynamic-panel GMM at the German Kreis level (dependent variable: BpCR); two-step with Windmeijer-corrected errors, lagged BpCR instrumented by lags 2-3, collapsed.

Variables	Arellano-Bond	Blundell-Bond
BpCR _{t-1}	+0.8218*** (0.0722)	+0.8231*** (0.0490)
ln(Urban Density)	-0.0498*** (0.0111)	-0.0004 (0.0006)
Urban Pop. Share	+0.0048 (0.0161)	-0.0004 (0.0006)
ln(GDP per capita)	-0.0051** (0.0023)	-0.0002 (0.0004)
Internal net migration	+0.0000 (0.0000)	-0.0000 (0.0000)
Sargan (p)	0.09	0.00
AR(2) (p)	0.36	0.32

*** p<0.01, ** p<0.05, * p<0.1.

Windmeijer-corrected robust standard errors in parentheses.

Three points emerge. First, per-capita land use is **strongly path-dependent** at the district level too: the lagged dependent variable is large and significant ($\rho \approx 0.82$), echoing the persistence found across countries. Second, the **income-densification** result holds: the difference-GMM (Arellano-Bond)

estimate keeps the negative, significant sign of the static model, in line with the national finding. Third, **internal net migration** is not significant within districts in this short panel; its clearer association with densification is cross-sectional, with the districts that attract internal migrants being the denser ones.

4 Caveats

The case study matches present-day GAUL districts to INKAR by name, so a minority of districts are dropped where the match is ambiguous; the controls limit the window to 1995–2020; and the GMM diagnostics, acceptable for the Arellano-Bond difference estimator, are weaker for the system estimator at this fine, short panel. The results should therefore be read as a corroborating local illustration rather than a stand-alone causal analysis. With those caveats, the case study reproduces the paper’s two central messages sub-nationally: LCRPGR is unusable where population growth is near zero while BpCR recovers an interpretable income-densification relationship; and it shows that the densifying role of in-migration also holds for the internal flows the national analysis cannot see.